ECE 315

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Lecture 29 :

- Analogue/Digital conversion
- ADCs, DACs

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Last Lecture

- Interrupts and Events

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This Lecture

- Analogue/Digital conversion
- ADCs, DACs

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Learning Objectives

- Distinguish between analogue and digital signals
- Explain the motivations for analogue-to-digital and digital-to-analogue conversion
- Describe the fundamentals of sampling theory
  - Apply the Nyquist–Shannon sampling theorem
- Identify practical limitations of real ADC systems, including trade-offs between
  - sampling rate and resolution, quantization levels, noise, and
  - embedded system constraints such as ADC speed, DMA bandwidth, & CPU load
- Explain why anti-aliasing and reconstruction filters are needed

Analogue and Digital Signals

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This is our main topic for until past Reading Week.

Why do we care about Analogue vs Digital?

- The world is for practical purposes analogue in nature
- Yes, there are analogue computers, but they are difficult to build and even more difficult to program
- Digital signal processing and by extension digital computers are so much easier to deal with
- Digital techniques, computing scales extraordinarily well, both in capability and cost
  - Many embedded systems are actually overkill in terms of what the microprocessor could do versus what it is asked to do

The World is Analogue†

- Until we go down to the level of quantum mechanics and quanta, the world is comprised of continua of physical phenomena
- We experience the world on a sliding scale
  - Things are louder or softer, brighter or darker, more red or less red, and so on

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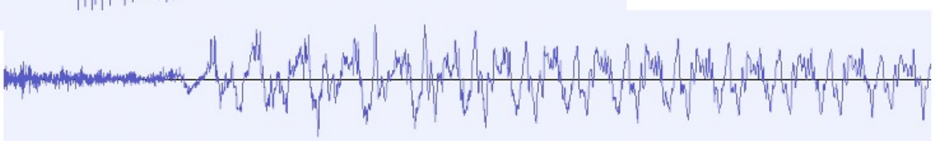
† …or if you live in the U.S., "analog"

The World is Analogue†

- The world is comprised of continua of physical phenomena
- We experience the world on a sliding scale
  - Things are louder or softer, brighter or darker, more red or less red, and so on
- Analogue signals take on values within some range



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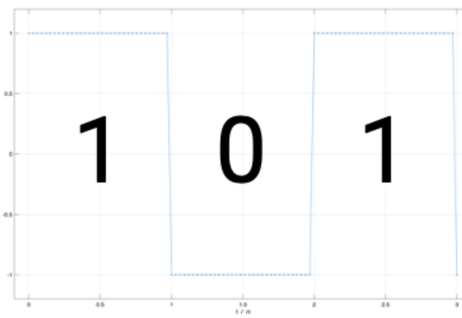
Digital Signals

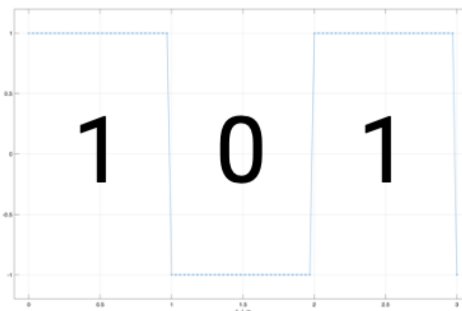
- Are either “ON” or “OFF”
- For digital systems it’s usual to say 0 or 1



Digital Signals cont'd

- Are either "ON" or "OFF"
- For digital systems it's usual to say 0 or 1
- The actual signals can be voltage, current, pressure, temperature, etc.

Signal	ON	OFF
Voltage	+5 V	0 V
Voltage	-12 V	12 V
Pressure (vacuum)	3 kPa	101.3 kPa



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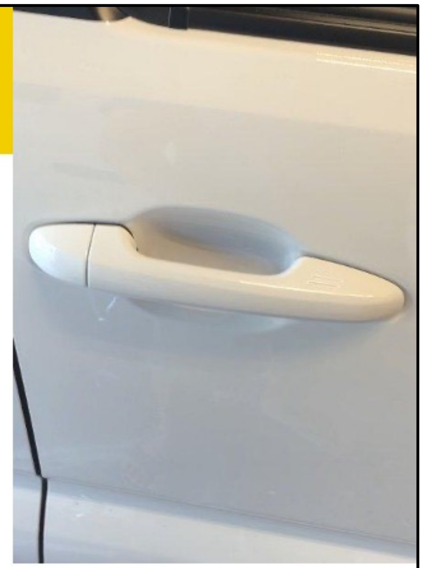
Analogue and Digital Signals

- My truck has a keyless lock.
- If my keys are in my pocket, I can unlock the truck by touching the door handle
- The sensor has to detect the key via wireless signals, and know my proximity
  - Both are analogue signals
- It will then change the door state to either locked or unlocked (0 or 1)



The wireless data signal, which is analogue between the transmitter and receiver, starts as digital signal at the transmitter, is converted to analogue and transmitted, is received and then converted to digital

The World is Analogue – cont'd

- The problem is noise and distortion
- When added, it is difficult if not impossible to remove
- It degrades what we experience, measure, and hence what we can do with signals

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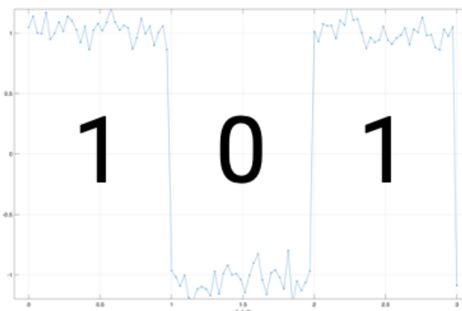
Digital Signals and noise

- If noise is added to a digital signal, as long as it doesn't exceed a certain threshold, we can still get the signal with perfect fidelity
- Advanced digital systems incorporate digital error detection and correction algorithms that will repair errors should they occur
  - First found widespread use in Compact Discs using the Reed-Solomon codes and algorithm



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Analogue Circuits

Advantages

- Most physical phenomena of interest are analogue
- Transducers are simple
- Potentially high precision

Disadvantages

- Behaviour of analog components is subject to drift distortion, noise, offsets, etc.
- Errors in analogue signals accumulate during processing, transmission, and storage
- Only relatively simple signal processing is practical for most applications

Digital Circuits

Advantages

- a digital signal is easily restored
- signal accuracy degrades very little, if at all, during processing, transmission and storage
- digital components are cheap, reliable and low-power
- digital signal processing can be highly sophisticated using special-purpose hardware or programmable digital computers

Disadvantages

- signal precision is limited by the number of bits used to encode each sample
- analog-to-digital converters and digital-to-analogue converters are required to interface a digital system with real-world analog signals

Sampling Theory Refresher

- Sampling converts a continuous-time signal $x(t)$ into a discrete-time sequence $x[n]$
- Performed by measuring the signal at regular intervals:

$$x[n] = x(nT_s)$$

where

  - T_s is the sample period in seconds, and $f_s = \frac{1}{T_s}$ is the sampling frequency (Hz)

- Common examples:
  - ADCs sampling sensor voltages
  - Audio signals sampled at 44.1 kHz or 48 kHz

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Nyquist-Shannon Sampling Theorem

- A band-limited signal with maximum frequency f_{max} can be perfectly reconstructed if

$$f_s \geq 2f_{max}$$

- If sampling is too slow aliasing occurs
  - high-frequency components appear as lower frequencies (said to ”wrap”)
- In practical applications, we need anti-aliasing filters that ensure

$$\frac{f_s}{2} \geq f_{max}$$

- Rule of thumb
  - Sample at 2–10 times f_{max} for margin and filter roll-off
- Special case is if a signal is bandlimited and bandpass in nature
  - Outside the scope of this course

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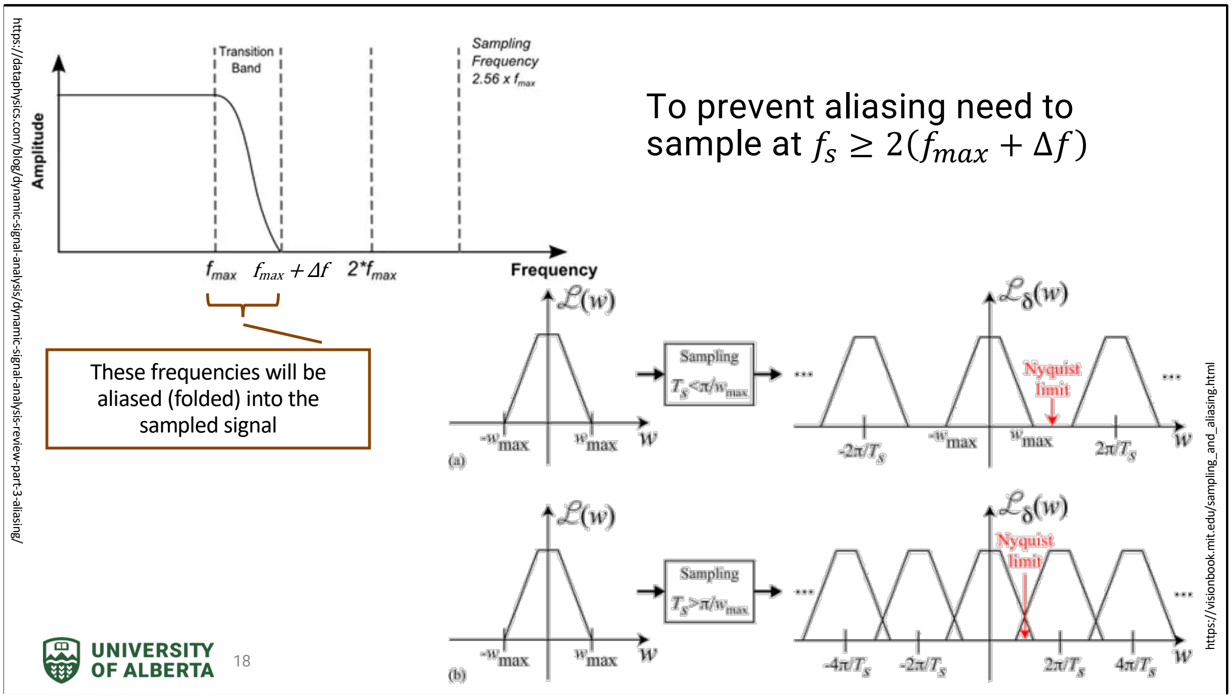

The top left figure shows the spectrum of a realistic, implementable anti-aliasing filter. There is significant frequency content greater than fmax that will be aliased, folded into the sampled signal. This distorts the desired signals.

The answer is to over-sample, sample at more than twice the Nyquist frequency.

Practical Considerations

- Aliasing:
  - Different continuous frequencies become indistinguishable after sampling
  - Once aliased, information is irreversibly lost
- Common mitigation strategies:
  - Analogue low-pass (anti-alias) filtering
  - Oversampling
  - Digital decimation after filtering
- In embedded systems:
  - Sampling rate often constrained by:
    - ADC speed
    - DMA bandwidth
    - CPU processing time

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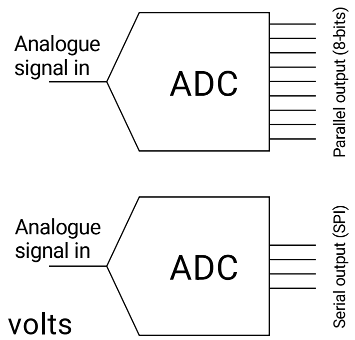
Analogue-to-Digital Convertor (ADC)

- An ADC takes an input analogue signal, samples it, and outputs the samples as digital words
- In general, the faster the ADC samples, the fewer the number of bit out
- An N-bit ADC can quantize the input signal to 2^N levels
  - Usually, they represent values that are symmetric about `0` volts
  - If the maximum voltage swing is ±X volts, the digital output range is approximately $[-2^{N-1}, 2^{N-1}]$ volts



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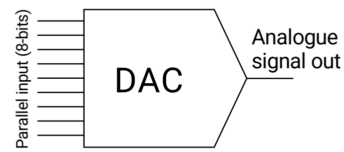
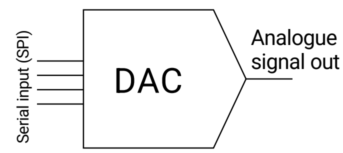
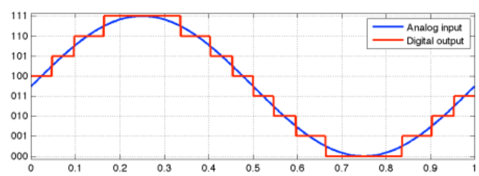
Digital-to-Analogue Convertor (DAC)

- A DAC converts digital values to analogue signals
- In general, the faster the ADC samples, the fewer the number of input bits
- The more input bits, the higher the fidelity of the output analogue signal
- Similar to the ADC, a filter is required for the output signal







https://dwellangle.wordpress.com/2008/05/21/quantization-in-adcs/

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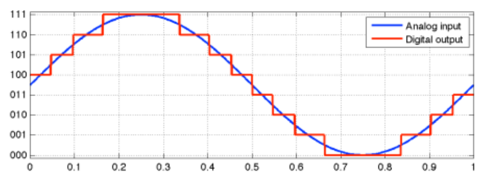
The output of the DAC, exaggerated here for a 3-bit DAC, contains a lot of high frequency components

Non-ideal DAC Operation



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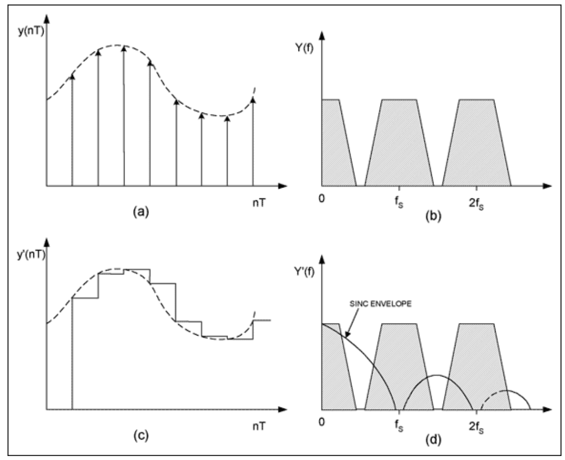

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https://www.analog.com/en/resources/technical-articles/equalizing-techniques-flatten-dac-frequency-response.html

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Ideally, the reconstructed signal is the original analogue signal multiplied by a sequence of delta functions. This leads to the spectrum of the original signal repeated every fs Hz.

Two things about that; first, we can't have a continuum of amplitudes, and second we can't make a delta function.

We approximate the desired signal by a discrete convolution with a narrow rectangle pulse.

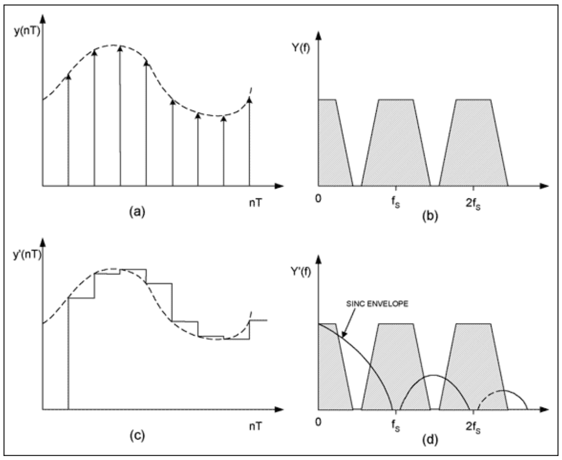
The result in the frequency domain is the multiplication of the fourier transform of the rectangle, a Sinc function, with the repeated spectrum.

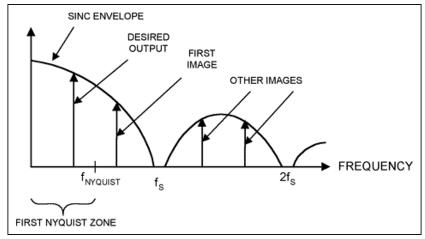
Similar to the aliased signal from the ADC operation, the output of the DAC will contain unwanted content from the repeated signals.

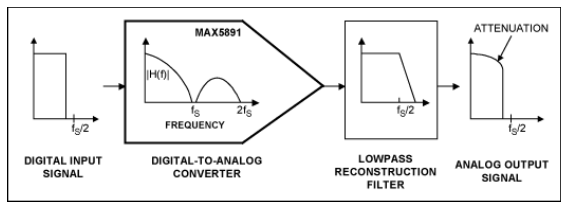
We need to low-pass filter the signal, ideally a "brick wall" filter with cutoff of fs/2

Multiplication in the time domain is the same as convolution in the frequency domain. We see this clearly in figure d) where the

Non-ideal DAC Operation







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https://www.analog.com/en/resources/technical-articles/equalizing-techniques-flatten-dac-frequency-response.html

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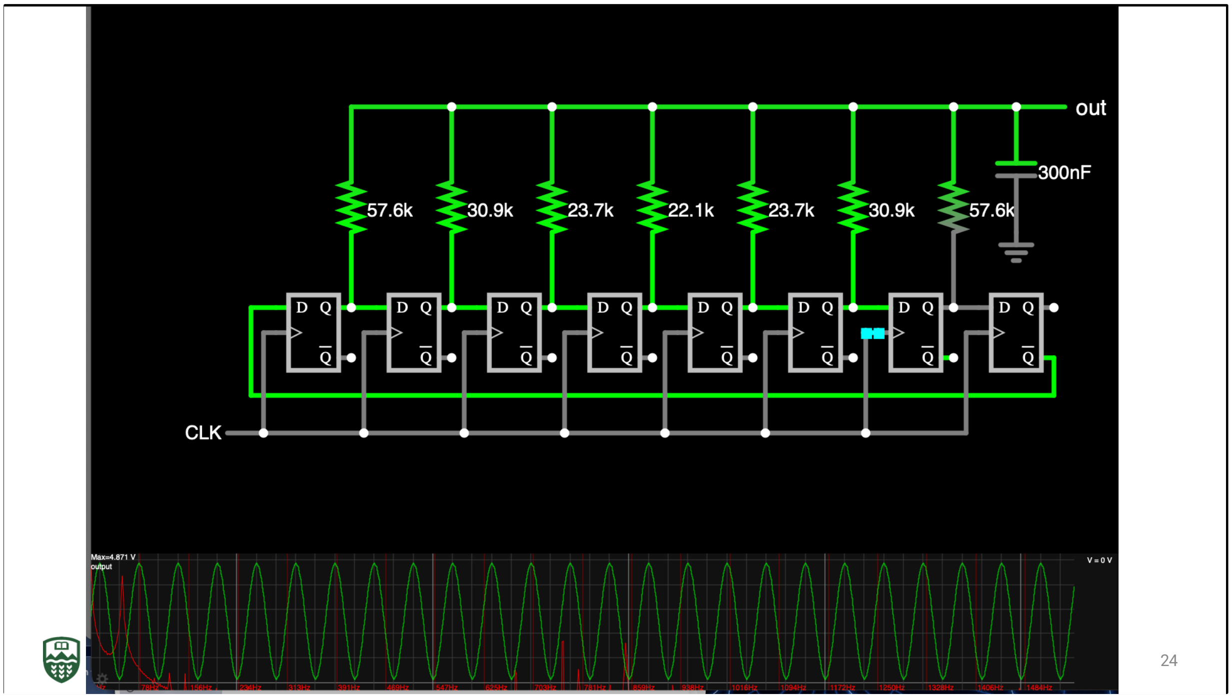
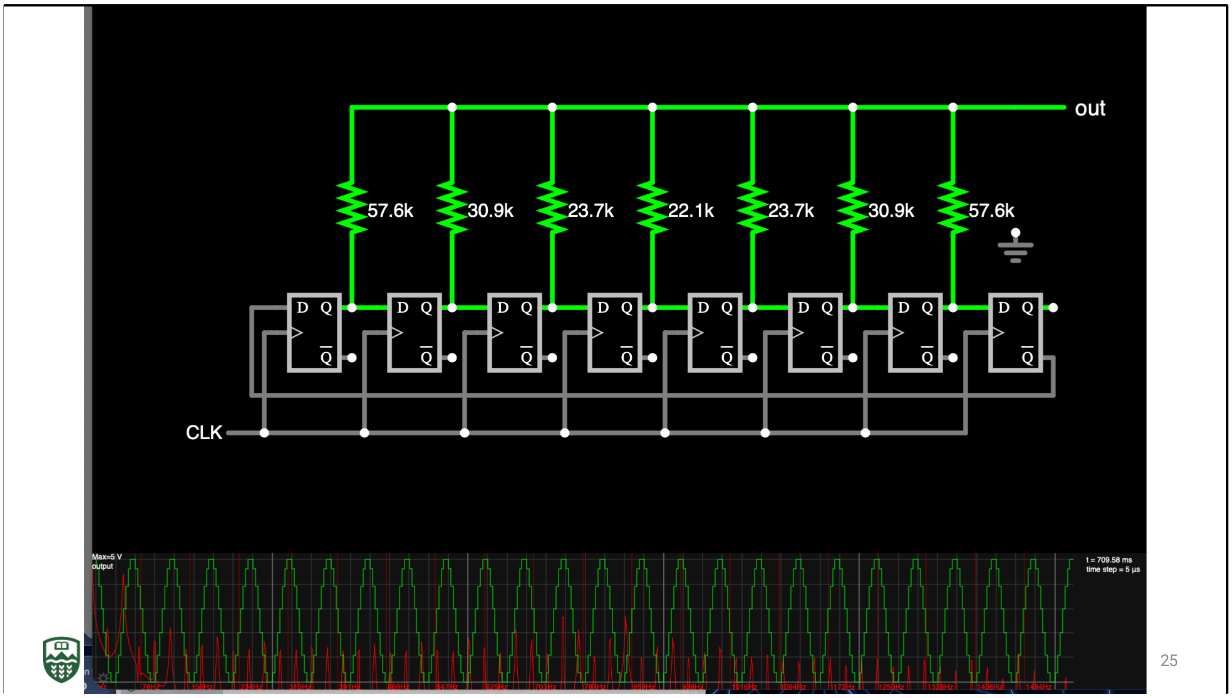
Here we see the practical signal processing chain with the lowpass “reconstruction” filter following the DAC



This is a DAC that outputs a fixed sinusoid. The green plot is the time-domain waveform while the red plot is the log frequency spectrum. The fundamental frequency of about 50 Hz is easy to see.

The output capacitor acts as a lowpass filter.



This is a DAC that outputs a fixed sinusoid. The green plot is the time-domain waveform while the red plot is the log frequency spectrum. The fundamental frequency of about 50 Hz is easy to see.

The output capacitor has been removed and we see the quantized time-domain signal. The spectrum now contains tons of extra frequencies, repeated every 50 Hz.

Yuck!

Analogue-to-Analogue processing

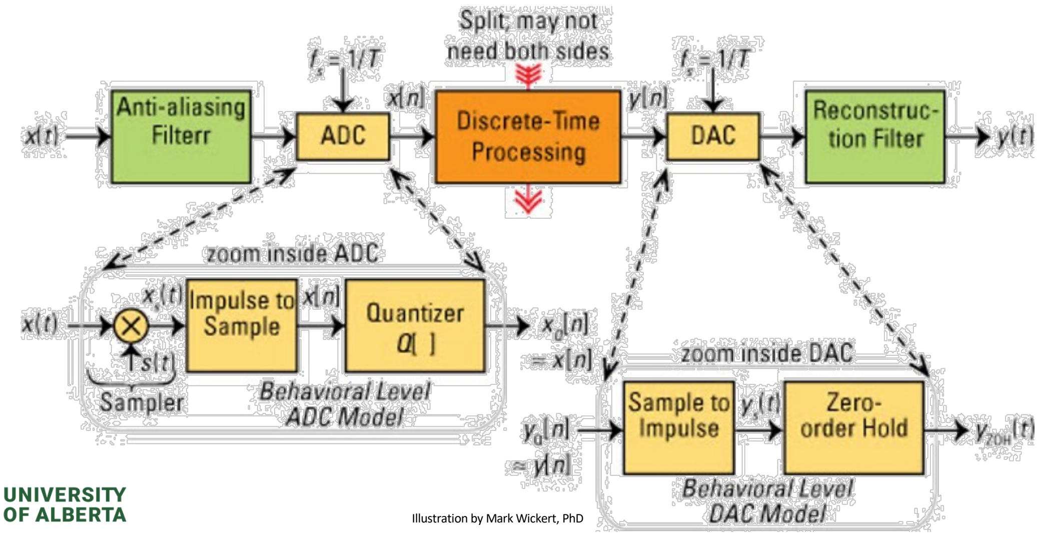


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Illustration by Mark Wickert, PhD

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This diagram illustrates practical blocks for both ADC and DAC operations

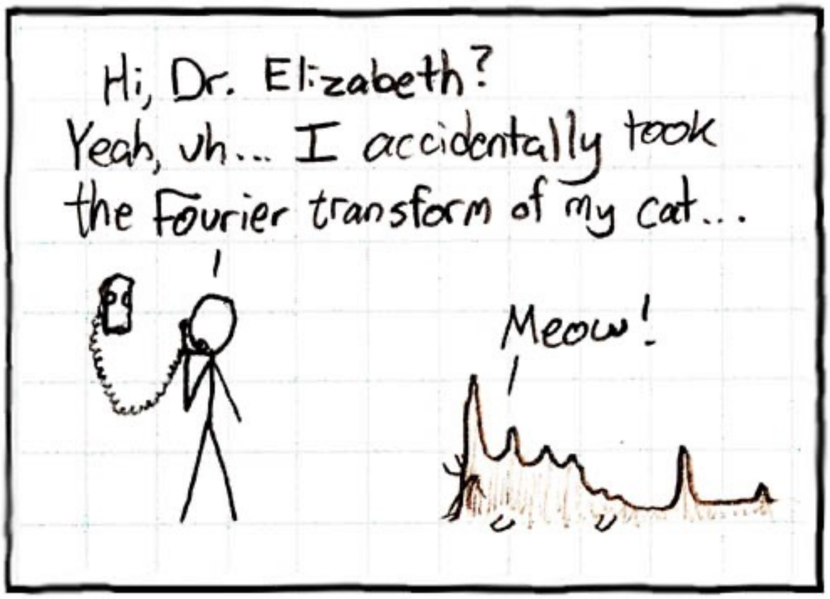
Summary

- The real world is inherently analogue, but digital systems dominate modern computing due to their robustness, scalability, and ease of processing; ADCs and DACs are needed
- Sampling theory describes how continuous-time signals are converted into discrete-time sequences using a fixed sampling frequency
- The Nyquist–Shannon sampling theorem defines the minimum sampling rate required to avoid aliasing; insufficient sampling causes irreversible distortion of digital signals
- Practical ADC systems employ band-limiting filters for input signal to reduce unwanted aliased signals
- Practical ADC systems involve trade-offs between sampling rate, resolution, noise, and embedded constraints such as ADC speed, DMA bandwidth, and CPU capacity
- DACs reconstruct analogue signals from digital values but introduce high-frequency artifacts, making reconstruction filtering essential for producing clean analogue outputs

Next Lecture

- Motion control
- Stepper and DC motors
- HW controllers, H-bridge

Questions



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https://xkcd.com/26/